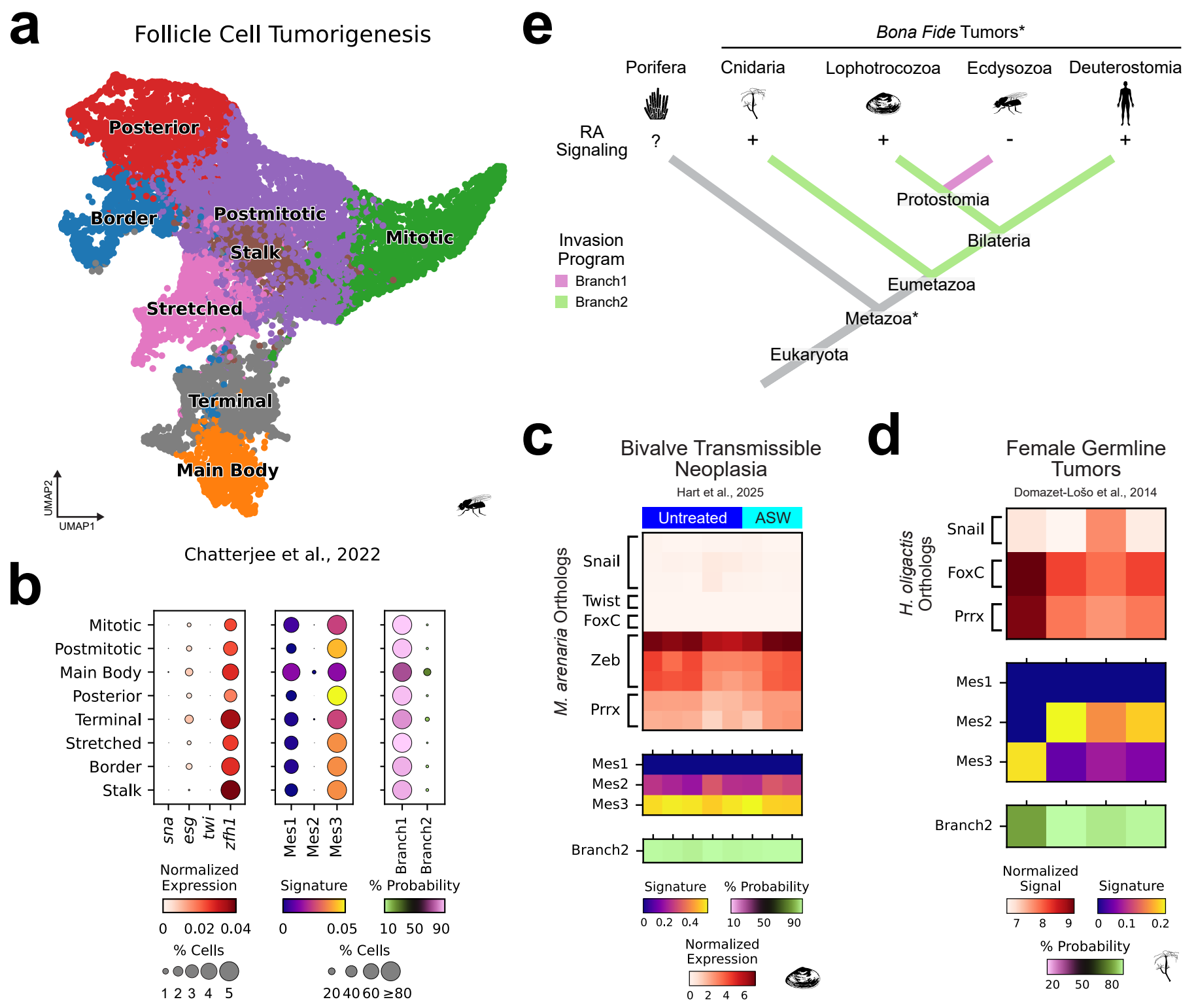

a
Follicle Cell Tumorigenesis
Posterior
Border
Postmitotic
Stalk
Mitotic
Stretched
Terminal
Main Body
UMAP2
UMAP1
Chatterjee et al., 2022
b
Mitotic
Postmitotic
Main Body
Posterior
Terminal
Stretched
Border
Stalk
sna
esg
twi
zfh1
Mes1
Mes2
Mes3
Branch1
Branch2
Normalized Expression
0 0.02 0.04
Signature
0 0.05
% Probability
10 50 90
% Cells
1 2 3 4 5
% Cells
20 40 60 ≥80
e
Bona Fide Tumors*
Porifera
Cnidaria
Lophotrocozoa
Ecdysozoa
Deuterostomia
RA Signaling
? + + - +
Invasion Program
Branch1
Branch2
Protostomia
Bilateria
Eumetazoa
Metazoa*
Eukaryota
c
Bivalve Transmissible Neoplasia
Hart et al., 2025
Untreated
ASW
M. arenaria Orthologs
Snail
Twist
FoxC
Zeb
Prrx
Mes1
Mes2
Mes3
Branch2
Signature
0 0.2 0.4
% Probability
10 50 90
Normalized Expression
0 2 4 6
d
Female Germline Tumors
Domazet-Lošo et al., 2014
H. oligactis Orthologs
Snail
FoxC
Prrx
Mes1
Mes2
Mes3
Branch2
Normalized Signal
7 8 9
Signature
0 0.1 0.2
% Probability
20 50 80